INCH-POUND

A-A-52214
January 9, 1997
SUPERSEDING
MIL-R-4286E
22 June 1972

COMMERCIAL ITEM DESCRIPTION

RIVETER, PNEUMATIC, COMPRESSION TYPE, PORTABLE, WITH RIVET SETS

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE

1.1 Scope. This commercial item description (CID) covers portable, compression type, pneumatic riveters with rivet sets.

1.2 Intended use. The riveters are intended for general utility maintenance usage to perform various riveting operations in conjunction with sheet metal bench work, modifications, and installations.

2. CLASSIFICATION

2.1 Items. The portable, compression type, pneumatic riveter shall conform to the following types:

Type I - Alligator jaw type
1 1/2-inch reach

Type II - C Yoke compression riveter
2 1/8-inch reach

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development and Engineering Center, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3 SALIENT CHARACTERISTICS

3.1 Design and construction. The riveters shall be manually-actuated, pneumatically-operated tools, designed for operation on an air pressure of 90 to 100 pounds per square inch gage (psig). The riveters shall consist of a cylinder, a set of jaws, a throttle and a handle and shall accommodate rivet sets conforming to 3.4. The riveters shall be designed to safely contain and exert the riveting pressures listed in 3.2.1 and 3.2.2. As an option, riveters may be equipped with a safety throttle assembly. When tested as specified in 5.3.2 and 5.3.3 there shall be no failure of operation, fracture, deformation, loosening of parts, damage to the riveter or rivet sets.

3.1.1 Throttles. The type I and type II riveters shall be provided with a nonleakable throttle of the lever type for controlling the air supply and for regulating the speed of the riveter.

3.1.2 Handles. The type I riveter shall be of such form that hand grips can be secured on the cylinder housing. The type II riveter shall be provided with straight handles attached to the end of the riveter opposite the jaws.

3.1.3 Air inlet connections. The air inlet connections to the riveter shall be provided with 1/4-inch internal pipe thread, and the air inlet shall have a screen to exclude particles from entering air controlled valves.

3.1.4 Fastening devices. All fasteners shall be installed with means to prevent loosening.

3.1.5 Interchangeability. All parts shall be manufactured to dimensional tolerances that shall permit replacement without modification.

3.2 Riveters.

3.2.1 Type I riveter. The type I riveter shall have alligator jaws, with one jaw stationary and the other moveable. The jaws shall accommodate and retain the rivet sets during the riveting operation and permit hand removal and insertion of rivet sets. The riveter shall have a minimum reach capacity of 1 3/8 inches, not weigh more than 5 pounds and not be greater than 11 inches in length. The force capacity using air pressure of 90 pounds per square inch shall equal or exceed 3,000 pounds. The riveter shall be tested for reach and force capacity as specified in 5.3.3.

3.2.2 Type II riveter. The type II riveter shall be the "C" yoke type with a stationary jaw and a moving plunger. The jaws shall accommodate and retain the rivet sets during the riveting operation and permit hand removal and insertion of rivet sets. The riveter shall have a minimum reach capacity of 2 inches, not weigh more than 13 pounds and not be greater than 19 inches in length. The force capacity using air pressure of 90 pounds per square inch shall equal or exceed 6,000 pounds. The riveter shall be tested for reach and force capacity as specified in 5.3.3.

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3.3 Workmanship. The finished riveter shall be clean, free of scratches, sharp edges, deformities, burrs, slivers or visible evidence of rust, dirt and other extraneous materials. Welding, brazing and soldering shall not be utilized as a repair measure for defective parts. Welding, brazing and soldering shall be performed only on those operations included in the original fabrication of the riveter. Welds shall be clean, continuous, sound and free of flux, pits, scale, buildups, undercuts and shall be ground so as to blend with adjacent surfaces. External surfaces shall be smooth and all edges rounded or beveled.

3.4 Rivet sets. Unless otherwise specified (see 7.1) a group of rivet sets shall be supplied with each riveter. The group shall consist of cupped and flush rivet sets to drive solid universal head rivets made of aluminum or aluminum alloy, steel or carbon and steel corrosion resistant, and brass copper or nickel copper alloy. The rivet sets shall be made of steel and shall withstand the functional tests specified in 5.3.3. A noncorrosive container with a cover and latch, shall be supplied with each group of rivet sets.

3.4.1 Rivet sets for type I riveter. The type I riveter shall be furnished with six assorted rivet sets as follows:

Two flush rivet sets having a 3/8 inch "A" dimension head length.
One flush rivet set having a 1/4 inch "A" dimension head length.
One flush rivet set having a 3/16 inch "A" dimension head length.
One each cupped rivet set having a 3/8 inch "A" dimension head length for 3/32 and 1/8 inch shank rivets. The shank diameters of all the rivet sets shall be .187 inch (see figures 1 and 2).

3.4.2 Rivet sets for type II riveter. The type II riveter shall be furnished with eight assorted rivet sets as follows:

One flush rivet set having a 1/4 inch "A" dimension head length.
One flush rivet set having a 1/2 inch "A" dimension head length.
One each cupped rivet set having a 1/4 inch "A" dimension head length for 1/8, 5/32 and 3/16 inch shank rivets.
One each cupped rivet set having a 1/2 inch "A" dimension head length for 1/8, 5/32 and 3/16 inch shank rivets. The shank diameter of all the rivet sets shall be either .187 or .250 inch (see figures 1 and 2).

3.5 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified using conversion tables contained in the latest revision of ASTM E 380, and all other requirements of this CID are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be

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made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

4. REGULATORY REQUIREMENTS

4.1 Contractor-recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR). Recycled or reclaimed materials may be used in the construction of the equipment described herein. Recycled and reclaimed materials shall consist of previously-used material which has been reprocessed to become a source of new raw materials. Under no conditions or circumstances shall the contractor submit to the Government for acceptance reconditioned or rebuilt components as a part of the equipment described herein.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, shall conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product that has been sold in the commercial marketplace for over two years, or is the same product that has successfully been delivered to the Government on a previous contract or purchase order. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

The following market acceptability criteria is necessary to document the quality of the product to be provided under this CID.

The company must have sold 1,000 items meeting the requirements of this CID in the commercial or Government marketplace over the past 2 years.

5.2 Warranty. Unless otherwise specified in the contract, the manufacturer's standard commercial warranty shall apply.

5.3 Tests.

5.3.1 Examination for weight and overall length. Type I and type II riveters shall be weighed and measured for overall length to meet the maximum requirements specified in 3.2.1 and 3.2.2.

5.3.2 Operational test. Each riveter shall be connected to an air supply source delivering not less than 90 psig. The throttle shall be gradually depressed in a manner to permit the riveter to go from a static position to full speed. The riveter shall be continuously cycled at full speed for a

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minimum period of 5 minutes. After shutting off the air supply, the riveter shall be examined for any failure of operation, fracture, deformation due to vibration, or loosening of fasteners.

5.3.3 Functional test. Operating from a 90 psig air supply source, the riveter shall cold drive 25, 1/8-inch aluminum rivets and tightly rivet 2 pieces of sheet metal. The contractor shall furnish the aluminum rivets and predrilled sheet metal. The sheet metal shall be of appropriate thickness to accommodate test rivets of a grip length compatible with the rivet sets specified. After cutting off the air supply, the riveter and rivet sets shall be examined for any damage to the riveter jaws or rivet sets. Additionally, during this test, reach and force capacity as specified in 3.2.1 and 3.2.2 shall be verified.

6. PACKAGING

6.1 Packaging. Preservation, packing and marking requirements shall be as specified in the contract or purchase order (see 7.1).

7. NOTES

7.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents (if applicable):

a. Title, number and date of this CID
b. Type of item required (see 2.1)
c. Packaging requirements (see 6.1)

7.2 Sources of documents

7.2.1 Sources for non-government association documents.

ASTM STANDARDS:

E 380 - Use of the International System of Units (SI) (The Modernized Metric System)

(Application for copies of ASTM standards should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959).

7.3 National Stock Numbers (NSN). The NSNs assigned to the pneumatic riveters covered by this commerical item description are:

Type I - 5130-00-241-3496
Type II - 5130-00-241-3497

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MILITARY INTERESTS:

Custodians
Army - GL
Air Force - 99

Review Activities
Army - AV
Navy - CG
Air Force - 84

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

PREPARING ACTIVITY:

Army-GL

Project 5130-0393

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RIVET SETS

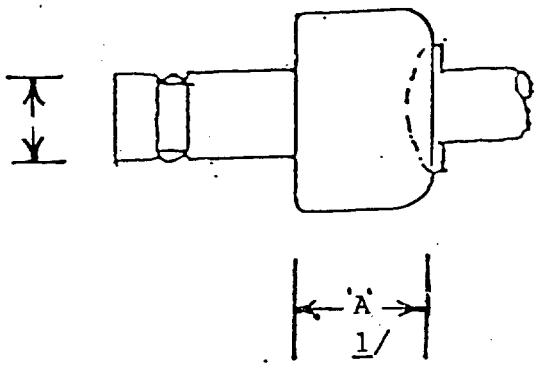


FIGURE 1. CUPPED SET DESIGN

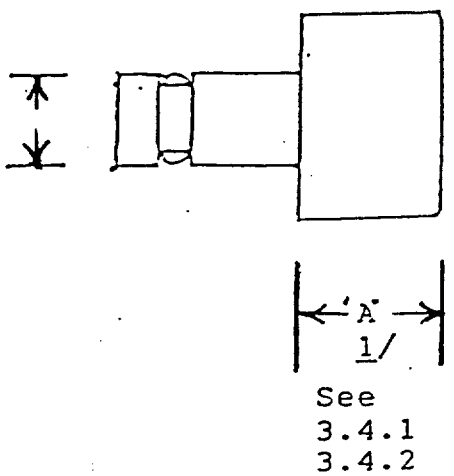


FIGURE 2. FLUSH OR FLAT SET DESIGN

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